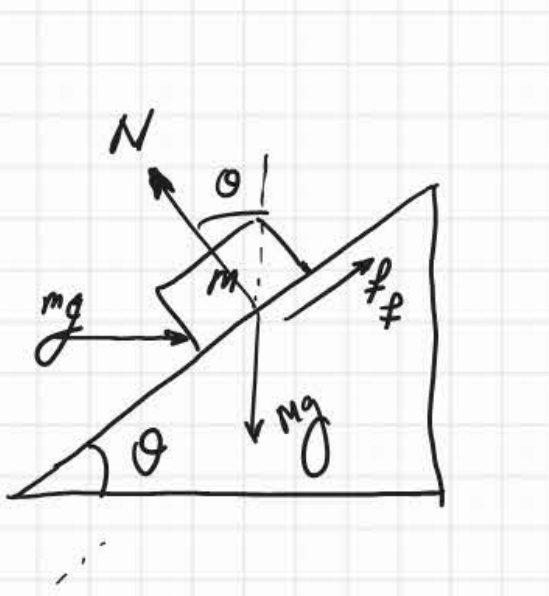
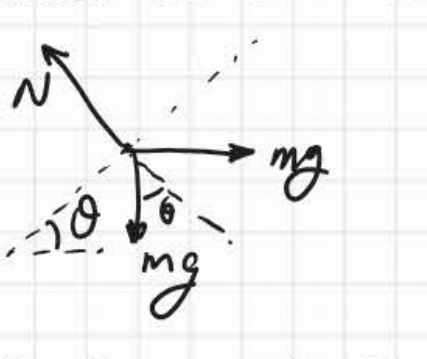
Problem 1.

Static setup $\Rightarrow \Sigma F = 0$





(a) $N = ?\ ;\ F_f = ?$

Normal force is the reaction to a compressoin force.

Two compression forces are involved:

$$N = mg\cos\theta + mg\sin\theta$$

F_f acts only along the surface: $F_f = mg\sin\theta - mg\cos\theta$

when is N maximum? $\frac{dN}{d\theta} = mg(\cos\theta - \sin\theta) = 0 \Rightarrow \boxed{\theta = \frac{\pi}{4}}$

(b) Static requires $|F_f| \le \mu N \Rightarrow mg|\sin\theta - \cos\theta| \le mg\mu(\sin\theta + \cos\theta)$

If $\sin\theta \geq \cos\theta$: $(1-\mu)\sin\theta \le (1+\mu)\cos\theta$ ①

$$\Rightarrow \tan\theta \le \frac{1+\mu}{1-\mu} \quad ; \quad \theta \geq 45$$

If $\sin\theta \le \cos\theta$: $mg(\cos\theta - \sin\theta) \le mg\mu\sin\theta + mg\mu\cos\theta$

$$\Rightarrow \cos\theta(1-\mu) \le \sin\theta(1+\mu)$$

$$\Rightarrow \tan\theta \geq \frac{1-\mu}{1+\mu} \quad \theta \le 45$$

$$\Rightarrow \boxed{\frac{1-\mu}{1+\mu} \le \tan\theta \le \frac{1+\mu}{1-\mu}}$$

Problem②:

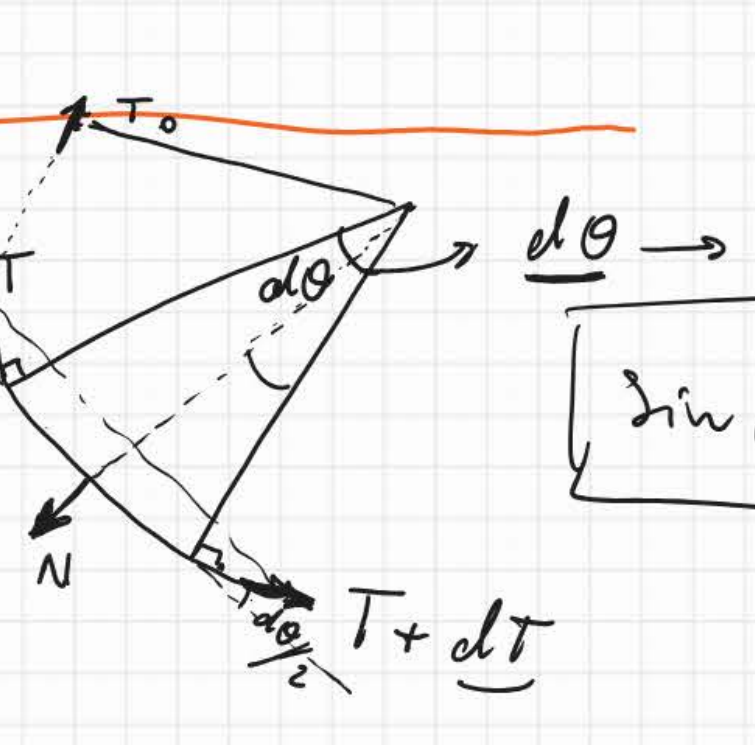
known: μ : coef. of static friction

T_0 : Tension

unknown: T_{max} while in static setup?

✓ $F_f \le \mu N \Rightarrow F_{f\,max} = \mu N$

✓ $\Sigma \vec{F} = 0$



$d\theta \to 0$

$\boxed{\sin d\theta \to d\theta}$

In the radial direction: $T\sin\frac{d\theta}{2} + (T + dT)\sin\frac{d\theta}{2} = N$

$$\Rightarrow N = 2T\sin\frac{d\theta}{2} + \underbrace{dT\sin\frac{d\theta}{2}} \simeq 2T\left(\frac{d\theta}{2}\right) = \boxed{T\,d\theta}$$

In the parallel direction: $\cancel{T} + dT \le T + F_f = T + \mu N = \cancel{T} + \mu T\,d\theta$

$$\Rightarrow \underset{\smile}{dT \le \mu T\,d\theta} \Rightarrow \int_{T_0}^{T_{max}} \frac{dT}{T} \le \int_0^{\theta} \mu\, d\theta$$

$$\Rightarrow \ln T \Big|_{T_0}^{T_{max}} \le \mu\theta \Rightarrow \ln T_{max} \le \mu\theta + \ln T_0$$

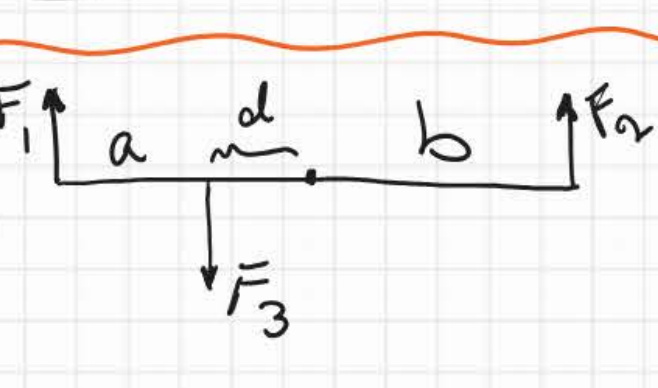
$$\Rightarrow \boxed{T_{max} \le T_0 e^{\mu\theta}}$$ in static configuration

Problem 3.

known: $F_3 = F_2 + F_1$

unknown: prove $F_3 a = F_2(a+b)$

eqn: $\Sigma F = 0$, $\Sigma \tau = 0$



$\Sigma F = 0 \Rightarrow F_3 = F_2 + F_1$; $\Sigma \tau = 0 \Rightarrow F_3 d + F_2(b-d) = F_1(a+d)$

$$d = \frac{b+a}{2} - a = \frac{b-a}{2}$$

$$F_3\left(\frac{b-a}{2}\right) + F_2\left(b - \frac{b-a}{2}\right) = F_1\left(a + \frac{b-a}{2}\right)$$

$$\Rightarrow F_3 \tfrac{b}{2} - F_3\tfrac{a}{2} + F_2 b - F_2\tfrac{b}{2} + F_2\tfrac{a}{2} - F_1 a - F_1\tfrac{b}{2} + F_1\tfrac{a}{2} = 0$$

$F_1 = F_3 - F_2$ $\quad F_2 \tfrac{b}{2}$

$$\Rightarrow \cancel{F_3\tfrac{b}{2}} - \cancel{F_3\tfrac{a}{2}} + \cancel{F_2 b} - \cancel{F_2\tfrac{b}{2}} + \cancel{F_2\tfrac{a}{2}} - F_3 a + F_2 a - \cancel{F_3\tfrac{b}{2}} + F_2\tfrac{b}{2}$$

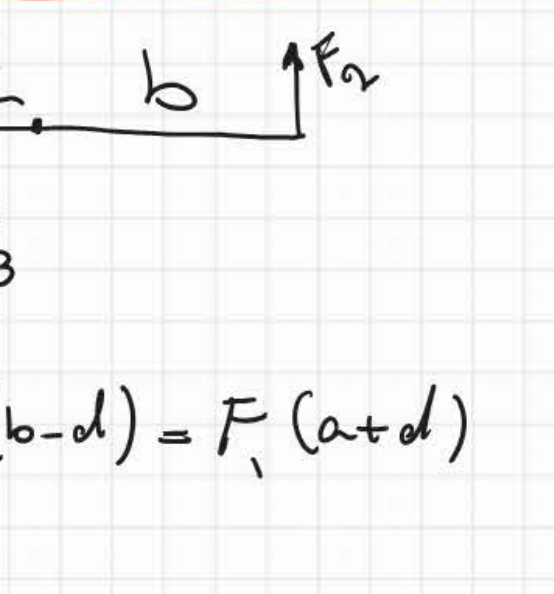
$$+ \cancel{F_3\tfrac{a}{2}} - \cancel{F_2\tfrac{a}{2}} = 0 \Rightarrow \boxed{F_2(a+b) = F_3 a}$$

Problem 4.

known: μ, $F_g = mg$, $l \equiv$ ladder length.

unknown: θ_{min} in static setup

eqns: $\Sigma F = 0$ $\Sigma \tau = 0$



$\Sigma \tau = 0 \Rightarrow N_2\sin\theta = N_1\cos\theta - F_f\sin\theta$

Let's take the pivot point to be the contact point C, because it contain the most number of forces, simplifying $\Sigma\tau = 0$.

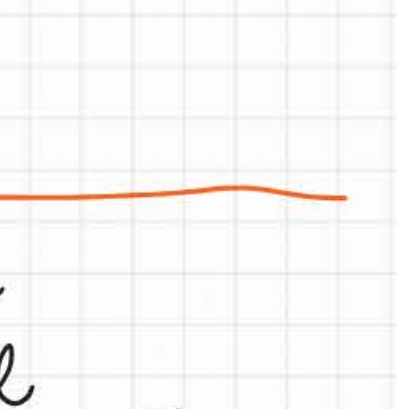
$$\Sigma \tau = 0\Big|_C \Rightarrow l N_2 \sin\theta = \frac{1}{2} l mg\cos\theta \Rightarrow \boxed{N_2 = \frac{mg}{2\tan\theta}}$$

$$\Sigma f_x = 0 \Rightarrow F_f = N_2 = \frac{mg}{2\tan\theta}$$

$$\left.\begin{array}{l} \Sigma F_y = 0 \Rightarrow N_1 = mg \\ \boxed{F_f \le \mu N_1} \end{array}\right\} \Rightarrow \frac{mg}{2\tan\theta} \le \mu mg \Rightarrow \boxed{\tan\theta \geq \frac{1}{2\mu}}$$

Problem 5.

known: m, μ, F, θ ; $\Sigma F = 0$, $\Sigma \tau = 0$



(a) unknown: $F_{min}(\theta) = ?$

$$\left.\begin{array}{l} \Sigma F_x = 0 \Rightarrow N = F\cos\theta \\ \Sigma F_y = 0 \Rightarrow F_{min}\sin\theta + \mu N = mg \end{array}\right\} \Rightarrow F_{min}(\sin\theta + \mu\cos\theta) = mg$$

$$\Rightarrow \boxed{F_{min} = \frac{mg}{\sin\theta + \mu\cos\theta}}$$

(b) $\underset{}{\operatorname{argmin}}\ F_{min}(\theta) = ?$

$$\rightsquigarrow \operatorname{argmax}\ \sin\theta + \mu\cos\theta = ? \quad \frac{d(\sin\theta + \mu\cos\theta)}{d\theta} = \cos\theta - \mu\sin\theta = 0$$

$$\Rightarrow \boxed{\mu = \operatorname{cotan}\theta} \text{ or } \tan\theta = \frac{1}{\mu} \ ; \ \cos^2\theta = \frac{1}{1+\tan^2\theta}$$

$$\Rightarrow F_{min} = \frac{mg}{\sqrt{1+\mu^2}}$$

(c) $F_{min} = \dfrac{mg}{\sin\theta + \mu\cos\theta} \to \infty \Rightarrow \sin\theta + \mu\cos\theta = 0$

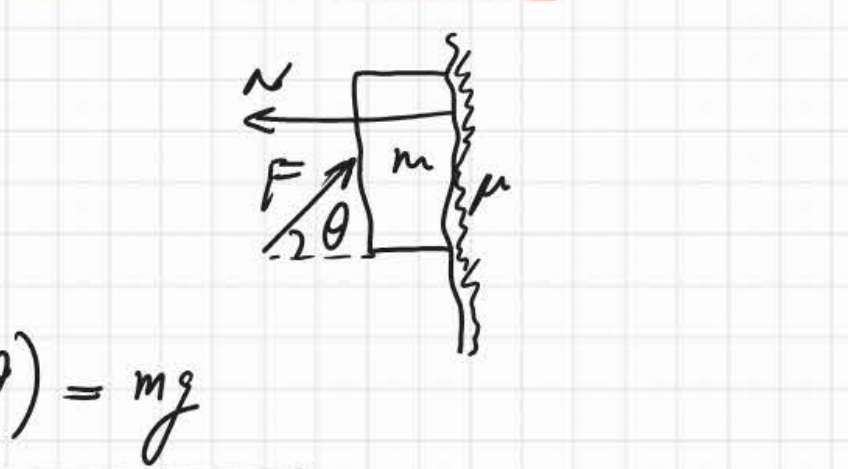
$$\Rightarrow \boxed{\tan\theta = -\mu}$$

Problem 6.

known: start–end at the same height, frictionless

unknown: prove the setup is static.

or $\Sigma F = 0$ along the tube



$$\underset{\smile}{dl} = \sqrt{dx^2 + f'^2 dx^2} = \sqrt{1 + f'^2(x)}\,dx \Rightarrow \boxed{dm = \rho\sqrt{1+f'^2}\,dx}$$

The gravity component along the tube:

$$dF_p = -dmg\sin\theta = -\rho\sqrt{1+f'^2}\,dx\, g\,\frac{f'dx}{dl} = \boxed{-g\rho f'(x)\,dx}$$

$$\Rightarrow F_p\Big|_a^b = \int_{x=a}^{x=b} dF_p = \int_a^b -g\rho f'(x)\,dx = -g\rho\left(\underbrace{f(b) - f(a)}_{=0}\right)$$

$$\boxed{= 0}$$

Problem 7

(a) The gravitational force downward on the disk is Mg, and the force upward is $2T$. These forces must balance, so

$$T = \frac{Mg}{2}. \tag{2.22}$$

We can find the normal force per unit length that the string applies to the disk in two ways.

FIRST METHOD: Let $N\,d\theta$ be the normal force on an arc of the disk that subtends an angle $d\theta$. Such an arc has length $R\,d\theta$, so N/R is the desired normal force per unit arclength. The tension in the string is the same throughout it, because the string is massless. So all points are equivalent, and hence N is constant, independent of θ. The upward component of the normal force is $N\,d\theta\cos\theta$, where θ is measured from the vertical (that is, $-\pi/2 \le \theta \le \pi/2$ here). Since the total upward force is Mg, we must have

$$\int_{-\pi/2}^{\pi/2} N\cos\theta\, d\theta = Mg. \tag{2.23}$$

The integral equals $2N$, so we have $N = Mg/2$. The normal force per unit length, N/R, therefore equals $Mg/2R$.

SECOND METHOD: Consider the normal force, $N\,d\theta$, on a small arc of the disk that subtends an angle $d\theta$. The tension forces on each end of the corresponding small piece of string almost cancel, but they don't exactly, because they point in slightly different directions. Their nonzero sum is what produces the normal force on the disk. From Fig. 2.43, we see that the two forces have a sum of $2T\sin(d\theta/2)$, directed "inward". Since $d\theta$ is small, we can use $\sin x \approx x$ to approximate this as $T\,d\theta$. Therefore, $N\,d\theta = T\,d\theta$, and so $N = T$. The normal force per unit arclength, N/R, therefore equals T/R. Using $T = Mg/2$ from Eq. (2.22), we arrive at $N/R = Mg/2R$.

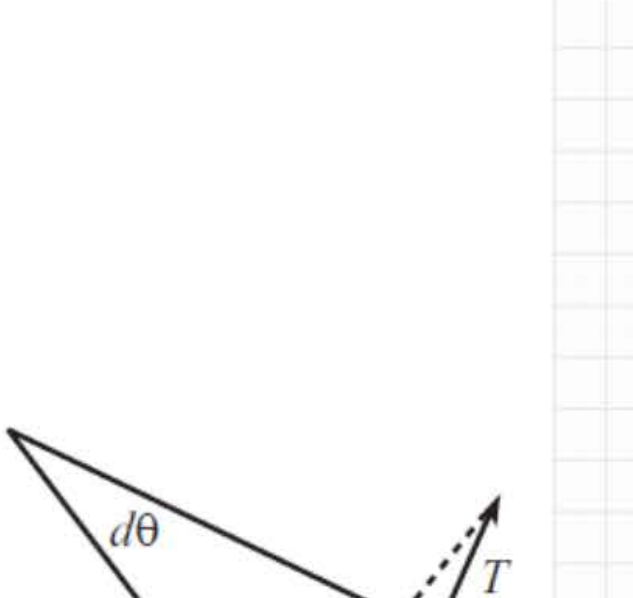


Fig. 2.43

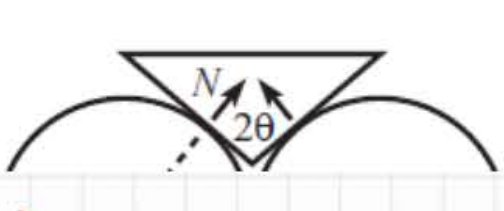
(b) Let $T(\theta)$ be the tension, as a function of θ, for $-\pi/2 \le \theta \le \pi/2$. T now depends on θ, because there is a tangential friction force. Most of the work for this problem was already done in the "Rope wrapped around a pole" example in Section 2.1. We'll simply invoke Eq. (2.7), which in the present language says[9]

$$T(\theta) \le T(0)e^{\mu\theta}. \tag{2.24}$$

Letting $\theta = \pi/2$, and using $T(\pi/2) = Mg/2$, gives $Mg/2 \le T(0)e^{\mu\pi/2}$. We therefore see that the tension at the bottom point must satisfy

$$T(0) \geq \frac{Mg}{2}e^{-\mu\pi/2}. \tag{2.25}$$

REMARK: This minimum value of $T(0)$ goes to $Mg/2$ as $\mu \to 0$, as it should. And it goes to zero as $\mu \to \infty$, as it should (imagine a very rough surface, so that the friction force from the rope near $\theta = \pi/2$ accounts for essentially all the weight). But interestingly, the tension at the bottom doesn't exactly equal zero, no matter now large μ is. Basically, the smaller T is, the smaller N is. But the smaller N is, the smaller the change in T is (because N determines the friction force). So T doesn't decrease much when it's small, and this results in it never being able to reach zero. ♣



Problem 8.

2.8. Hanging chain

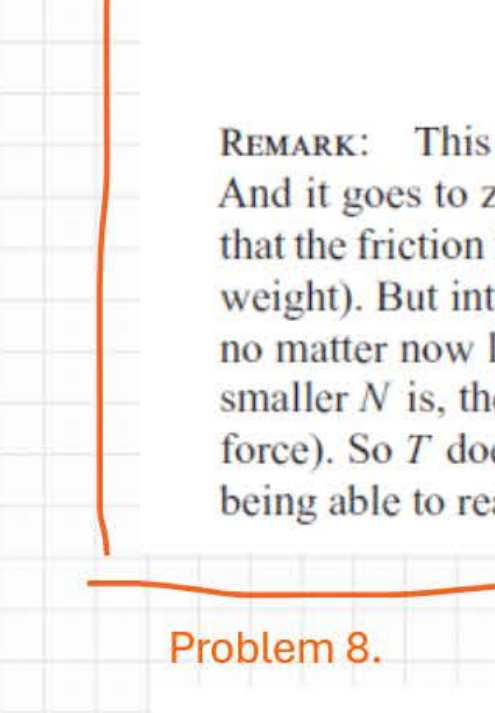


Fig. 2.47

(a) The key fact to note is that the horizontal component, T_x, of the tension is the same throughout the chain. This is true because the net horizontal force on any subpart of the chain must be zero. Label the constant value as $T_x \equiv C$.

Let the shape of the chain be described by the function $y(x)$. Since the tension points along the chain at all points (see Problem 2.12), its components satisfy $T_y/T_x = y'$, which gives $T_y = Cy'$. In other words, T_y is proportional to the slope of the chain.

Now consider a little piece of the chain, with endpoints at x and $x + dx$, as shown in Fig. 2.47. The difference in the T_y values at the endpoints is what balances the weight of the little piece, $(dm)g$. The length of the piece is $ds = dx\sqrt{1+y'^2}$, so if ρ is the density, we have

$$dT_y = (\rho\, ds)g = \rho g\, dx\sqrt{1+y'^2} \implies \frac{dT_y}{dx} = \rho g\sqrt{1+y'^2}. \tag{2.31}$$

Using the $T_y = Cy'$ result from above, this becomes $Cy'' = \rho g\sqrt{1+y'^2}$. Letting $z \equiv y'$, we can separate variables and integrate to obtain

$$\int \frac{dz}{\sqrt{1+z^2}} = \int \frac{\rho g\, dx}{C} \implies \sinh^{-1} z = \frac{\rho g x}{C} + A, \tag{2.32}$$

where A is a constant of integration. We can make this look a little cleaner if we define constants α and a such that $\alpha \equiv \rho g/C$ and $a \equiv A/\alpha$. We then obtain

$$\sinh^{-1} z = \alpha(x+a) \implies z = \sinh\alpha(x+a). \tag{2.33}$$

Recalling that $z \equiv dy/dx$, we can integrate again to obtain

$$y(x) = \frac{1}{\alpha}\cosh\alpha(x+a) + h. \tag{2.34}$$

The shape of the chain is therefore a hyperbolic cosine function. The constant h isn't too important, because it depends simply on where we pick the $y = 0$ height. Furthermore, we can eliminate the need for the constant a if we pick $x = 0$ to be where the lowest point of the chain is (or where it would be, in the case where the slope is always nonzero). In this case, using Eq. (2.34), we see that $y'(0) = 0$ implies $a = 0$, as desired. We then have (ignoring the constant h) the nice simple result,

$$y(x) = \frac{1}{\alpha}\cosh(\alpha x). \tag{2.35}$$

(b) The constant α can be determined from the locations of the endpoints and the length of the chain. As stated in the problem, the position of the chain may be described by giving (1) the horizontal distance d between the two endpoints, (2) the vertical distance λ between the two endpoints, and (3) the length ℓ of the chain, as shown in Fig. 2.48. Note that it isn't obvious what the horizontal distances between the ends and the minimum point (which we have chosen as the $x = 0$ point) are. If $\lambda = 0$, then these distances are $d/2$, by symmetry. But otherwise, they aren't so clear.

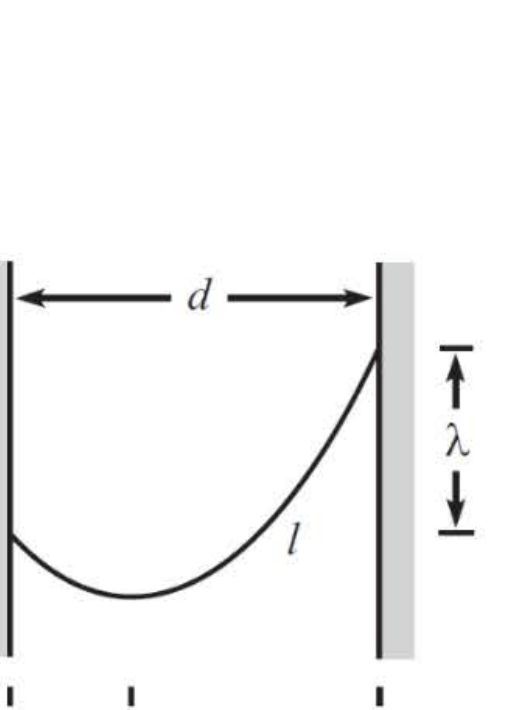


Fig. 2.48

If we let the left endpoint be located at $x = -x_0$, then the first of the above three facts says that the right endpoint is located at $x = d - x_0$. We now have two unknowns, x_0 and α. The second fact tells us that (we'll take the right end to be higher than the left end, without loss of generality)

$$y(d - x_0) - y(-x_0) = \lambda. \tag{2.36}$$

And the third fact gives, using Eq. (2.35),

$$\ell = \int_{-x_0}^{d-x_0} \sqrt{1+y'^2}\, dx = \frac{1}{\alpha}\sinh(\alpha x)\Big|_{-x_0}^{d-x_0}, \tag{2.37}$$

where we have used $(d/du)\cosh u = \sinh u$, and $1 + \sinh^2 u = \cosh^2 u$, and $\int \cosh u = \sinh u$. Writing out Eqs. (2.36) and (2.37) explicitly, we have

$$\begin{aligned} \cosh\big(\alpha(d - x_0)\big) - \cosh(-\alpha x_0) &= \alpha\lambda, \\ \sinh\big(\alpha(d - x_0)\big) - \sinh(-\alpha x_0) &= \alpha\ell. \end{aligned} \tag{2.38}$$

We can eliminate x_0 by taking the difference of the squares of these two equations. Using the hyperbolic identities $\cosh^2 u - \sinh^2 u = 1$ and $\cosh u\cosh v - \sinh u\sinh v = \cosh(u - v)$, we obtain

$$2\cosh(\alpha d) - 2 = \alpha^2(\ell^2 - \lambda^2). \tag{2.39}$$

This is the desired equation that determines α. Given d, λ, and ℓ, we can numerically solve for α. Using a "half-angle" formula, you can show that Eq. (2.39) may also be written as

$$2\sinh(\alpha d/2) = \alpha\sqrt{\ell^2 - \lambda^2}. \tag{2.40}$$

REMARK: Let's check a couple limits. If $\lambda = 0$ and $\ell = d$ (that is, the chain forms a horizontal straight line), then Eq. (2.40) becomes $2\sinh(\alpha d/2) = \alpha d$. The solution to this is $\alpha = 0$, which does indeed correspond to a horizontal straight line, because for small α, we can use $\cosh\epsilon \approx 1 + \epsilon^2/2$ to say that the $y(x)$ in Eq. (2.35) behaves like $\alpha x^2/2$ (up to an additive constant), which varies slowly with x for small α. Another limit is where ℓ is much larger than both d and λ. In this case, Eq. (2.40) becomes $2\sinh(\alpha d/2) \approx \alpha\ell$. The solution to this is a large α (or more precisely, $\alpha \gg 1/d$), which corresponds to a "droopy" chain, because the $y(x)$ in Eq. (2.35) varies rapidly with x for large α. ♣

Problem 9)

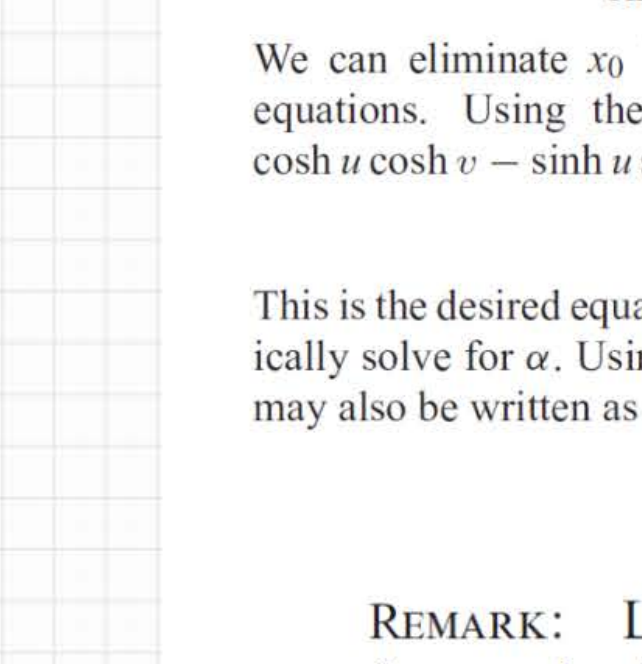


Fig. 2.49

We must first find the mass of the chain by calculating its length. Then we must determine the slope of the chain at the supports, so we can find the components of the force there. Using the given information, $y(x) = (1/\alpha)\cosh(\alpha x)$, the slope of the chain as a function of x is

$$y' = \frac{d}{dx}\left(\frac{1}{\alpha}\cosh(\alpha x)\right) = \sinh(\alpha x). \tag{2.41}$$

The total length is therefore (using $1 + \sinh^2 z = \cosh^2 z$)

$$\ell = \int_{-d}^{d}\sqrt{1+y'^2}\, dx = \int_{-d}^{d}\cosh(\alpha x) = \frac{2}{\alpha}\sinh(\alpha d). \tag{2.42}$$

The weight of the rope is $W = \rho\ell g$, where ρ is the mass per unit length. Each support applies a vertical force of $W/2$. So this equals $F\sin\theta$, where F is the magnitude of the force at each support, and θ is the angle it makes with the horizontal. Since $\tan\theta = y'(d) = \sinh(\alpha d)$, we see from Fig. 2.49 that $\sin\theta = \tanh(\alpha d)$. Therefore,

$$F = \frac{1}{\sin\theta}\cdot\frac{W}{2} = \frac{1}{\tanh(\alpha d)}\cdot\frac{\rho g\sinh(\alpha d)}{\alpha} = \frac{\rho g}{\alpha}\cosh(\alpha d). \tag{2.43}$$

Taking the derivative of this (as a function of α), and setting the result equal to zero to find the minimum, gives $\tanh(\alpha d) = 1/(\alpha d)$. This must be solved numerically. The result is

$$\alpha d \approx 1.1997 \equiv \eta. \tag{2.44}$$

So α is given by $\alpha = \eta/d$, and the shape of the chain that requires the minimum F is thus

$$y(x) \approx \frac{d}{\eta}\cosh\left(\frac{\eta x}{d}\right). \tag{2.45}$$

From Eqs. (2.42) and (2.44), the length of the chain is $\ell = (2d/\eta)\sinh(\eta) \approx (2.52)d$. To further get an idea of what the chain looks like, we can calculate the ratio of the height, h, to the width, $2d$.

$$\frac{h}{2d} = \frac{y(d) - y(0)}{2d} = \frac{\cosh(\eta) - 1}{2\eta} \approx 0.338. \tag{2.46}$$

We can also calculate the angle of the rope at the supports, using $\tan\theta = \sinh(\alpha d)$. This gives $\tan\theta = \sinh\eta$, and so $\theta \approx 56.5°$.

REMARK: We can also ask what shape the chain should take in order to minimize the horizontal or vertical component of F. The vertical component, F_y, is simply half the weight, so we want the shortest possible chain, namely a horizontal one (which requires an infinite F). This corresponds to $\alpha = 0$. The horizontal component, F_x, equals $F\cos\theta$. From Fig. 2.49, we see that $\cos\theta = 1/\cosh(\alpha d)$. Therefore, Eq. (2.43) gives $F_x = \rho g/\alpha$. This goes to zero as $\alpha \to \infty$, which corresponds to a chain with infinite length, that is, a very "droopy" chain. ♣

Problem 10)

(a) Let F and T be the tensions at the wall and the lowest point, respectively. Looking at the y forces on half of the chain gives $F\cos\theta = (M/2)g$, and looking at the x forces gives $F\sin\theta = T$. These yield $T = (M/2)g\tan\theta$.

(b) The slope of the chain is $y' = \sinh\alpha x$, which is approximately αx for small x. Consider a small piece that goes from $-x$ to x. The weight is essentially $\rho(2x)g$. The upward component of the tensions at the two ends is essentially $2Ty' \approx 2T(\alpha x)$. Balancing these gives $T = \rho g/\alpha$.

Now let's find α. The length from the bottom, as a function of a general value of x, equals

$$\int_0^x \sqrt{1+y'^2}\, dx = \int_0^x \cosh\alpha x = (1/\alpha)\sinh\alpha x. \tag{11}$$

Therefore, $M/2 = (\rho/\alpha)\sinh\alpha x_0$, where x_0 is the location of the wall. But the slope at the wall is $\sinh\alpha_0 x = 1/\tan\theta$. So $M/2 = \rho/(\alpha\tan\theta) \implies \rho/\alpha = (M/2)\tan\theta$. Plugging this into the above T gives $T = (M/2)g\tan\theta$, in agreement with part (a).

Problem 11)

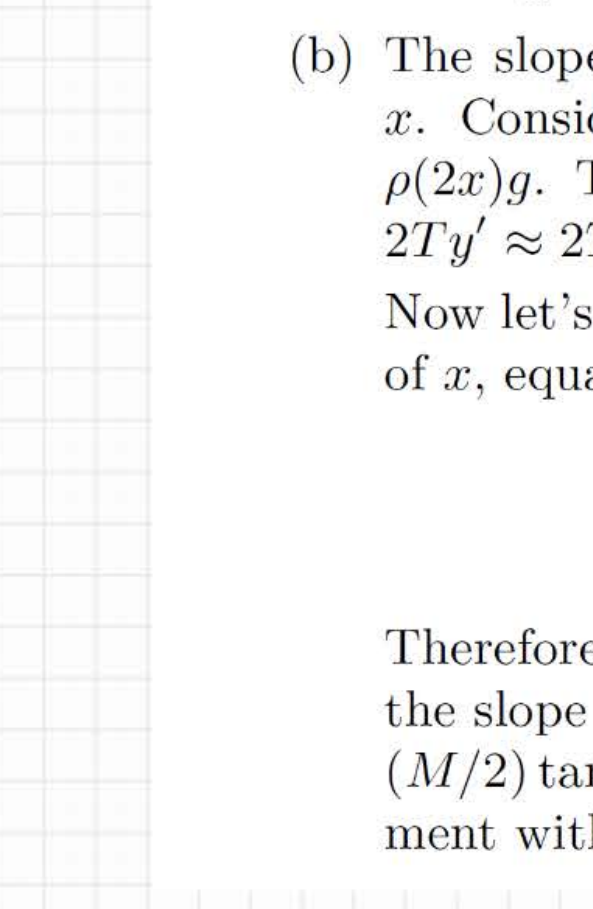


Figure 2

For convenience, let $d = \ell/2$, so the blocks have length $2d$. Then if $N = 1$, the answer is simply d.

If $N = 2$, then as far as the top block is concerned, the bottom block is simply the table from the $N = 1$ case. The top block can therefore hang out a distance d beyond the bottom block. The CM of the two-block system is a distance $d/2$ from the right end of the bottom block. In the cutoff case, the CM is right over the edge of the table, so this means that the bottom block can hang out a distance $d/2$ past the edge. The total overhang is therefore $d + d/2$.[1]

If $N = 3$, then as far as the top two blocks are concerned, the bottom block is simply the table from the $N = 2$ case. The top two blocks can therefore hang out a distance $d + d/2$ beyond the bottom block; see the first setup in Fig. 2. The CM of the entire system is a distance $d/3$ from the right end of the bottom block (because the top two blocks are effectively a point mass $2m$ sitting directly over the right end of the bottom block; see the second setup in Fig. 2). In the cutoff case, the CM of the entire system is right over the edge of the table, so this means that the bottom block can hang out a distance $d/3$ past the edge. The total overhang is therefore $d + d/2 + d/3$.

At this point (or perhaps after doing the $N = 4$ case) we can guess that the general answer is $d(1 + 1/2 + 1/3 + \cdots + 1/N)$. Let's prove this by induction. For a general N, we'll assume that the result is true for $N - 1$ and then prove that it is also true for N. As far as the top $N - 1$ blocks are concerned, the bottom block is simply the table from the $N - 1$ case. The top $N - 1$ blocks can therefore hang out a distance $d\big(1 + 1/2 + \cdots + 1/(N-1)\big)$ beyond the bottom block. The CM of the entire system is a distance d/N from the right end of the bottom block (because the top $N - 1$ blocks are effectively a point mass $(N - 1)m$ sitting directly over the right end of the bottom block). In the cutoff case, the CM of the entire system is right over the edge of the table, so this means that the bottom block can hang out a distance d/N past the edge. The total overhang L is therefore

$$L = d\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N-1} + \frac{1}{N}\right), \tag{18}$$

as we wanted to show. Since the result is true for $N = 1$, it is therefore true for all N.

For large N, this result behaves like $d\ln N$. So for $N \to \infty$, we can make the blocks hang out infinitely far. But the result grows very slowly with N, like a log. If we want the overhang to be L, then we need $d\ln N \approx L \implies N \approx e^{L/d}$, which grows very quickly with L.

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[1] Note that there is nothing to be gained by moving the top block a little to the left, with the hope that since the CM of the two-block system has moved to the left, both blocks can then be moved to the right, thereby possibly increasing the total overhang. If the top block is moved x to the left, then the CM of the two-block system moves only $x/2$ to the left. So after the system is moved to the right by this $x/2$, there is still a net loss of $x - x/2 = x/2$ in the overhang. Similar arguments hold for other subsystems in the more general cases below.

Problem 12)

Let's break up the forces into components parallel and perpendicular to the overhang. Let positive F_f point up along the overhang. Balancing the forces parallel and perpendicular to the overhang gives, respectively,

$$\begin{aligned} F_f &= Mg\sin\beta + Mg\cos\beta, \quad \text{and} \\ N &= Mg\sin\beta - Mg\cos\beta. \end{aligned} \tag{5}$$

N must be positive, so we immediately see that β must be at least 45° if there is any chance that the setup is static.

The coefficient μ tells us that $|F_f| \le \mu N$. Using Eq. (5), this inequality becomes

$$Mg(\sin\beta + \cos\beta) \le \mu Mg(\sin\beta - \cos\beta) \implies \frac{\mu+1}{\mu-1} \le \tan\beta. \tag{6}$$

We see that we must have $\mu > 1$ in order for there to exist any values of β that satisfy this inequality. If $\mu \to \infty$, then β can be as small as 45°, but it can't be any smaller.